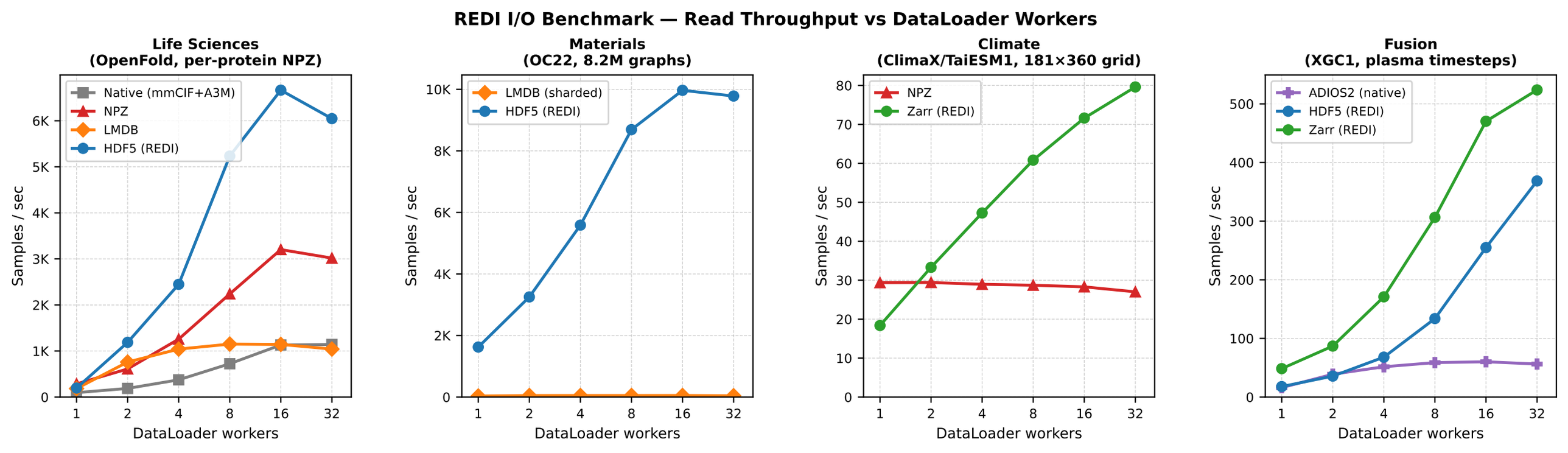
REDI I/O Benchmark — Read Throughput vs DataLoader Workers
Life Sciences
(OpenFold, per-protein NPZ)
Native (mmCIF+A3M)
NPZ
LMDB
HDF5 (REDI)
Samples / sec
DataLoader workers
Materials
(OC22, 8.2M graphs)
LMDB (sharded)
HDF5 (REDI)
Climate
(ClimaX/TaiESM1, 181×360 grid)
NPZ
Zarr (REDI)
Fusion
(XGC1, plasma timesteps)
ADIOS2 (native)
HDF5 (REDI)
Zarr (REDI)